

Model Deployment — API & Containerization Project

Goal

Wrap a trained machine learning model in a Flask API and containerize it using Docker.

Technologies Used

Python, Flask, Docker, NumPy, Scikit-learn

Project Structure

```
model_deployment_project/  
  app.py  
  Dockerfile  
  requirements.txt  
  README.md  
  curl_example.txt
```

Flask API Endpoint

POST /predict

Example Request

```
{ "features": [1,2,3,4] }
```

Example Response

```
{ "prediction": 1 }
```

Run Locally

```
pip install -r requirements.txt  
python app.py
```

Docker Commands

```
docker build -t ml-api .  
docker run -p 5000:5000 ml-api
```

Curl Example

```
curl -X POST http://127.0.0.1:5000/predict \  
-H "Content-Type: application/json" \  
-d "{\"features\": [1, 2, 3, 4]}"
```

Conclusion

This project demonstrates model deployment using Flask and Docker with API inference support.